

The probability-level lifting map is always surjective

An affirmative answer to the open question in arXiv:2402.04926

Autonomous solve-first investigation, lane 12

12 August 2026

Abstract

Oberguggenberger asks whether the probability-level evaluation map κ_1 for algebraic-dual-valued random functionals is surjective for every locally convex space. It is. The unrestricted map κ_0 is surjective by a Hamel-basis lifting, and every κ_0 -lift of a sequentially continuous map $E \rightarrow L^0(\Omega)$ is automatically continuous in probability. Thus restricting both domain and codomain to the probability-continuous level preserves surjectivity. The proof works for every probability space and every Hausdorff locally convex space, with no nuclearity, separability, completeness, or regularity assumption.

1 The question

Fix a probability space (Ω, Σ, P) and a Hausdorff locally convex space E . Give the algebraic dual E^* the cylinder sigma-algebra. Write $L^0(\Omega)$ for real measurable functions modulo almost-sure equality, with its usual topology of convergence in probability.

For a cylinder-measurable map $X : \Omega \rightarrow E^*$, define

$$(\kappa_0 X)(e) = [\omega \mapsto X(\omega)(e)], \quad e \in E.$$

The paper writes $L^0(\Omega : E^*)_{\text{prob}}$ for those X such that, whenever $e_n \rightarrow e$ in E ,

$$X(\cdot)(e_n) \longrightarrow X(\cdot)(e) \quad \text{in probability.}$$

Its map κ_1 is the restriction

$$\kappa_1 : L^0(\Omega : E^*)_{\text{prob}} \longrightarrow \mathcal{L}(E : L^0(\Omega))_{\text{prob}} = \mathcal{L}_c(E : L^0(\Omega)). \quad (1)$$

At the end of Section 7, the paper asks whether (1) is surjective in general [1, p. 17]. The exact passage is reproduced in Figure 1.

2 A general affirmative answer

Theorem 1. *For every probability space (Ω, Σ, P) and every Hausdorff locally convex space E , the map κ_1 in (1) is surjective.*

Proof. Let $Y \in \mathcal{L}_c(E : L^0(\Omega))$. Choose a Hamel basis $(u_\lambda)_{\lambda \in \Lambda}$ of E . For each λ , choose a measurable representative $y_\lambda : \Omega \rightarrow \mathbb{R}$ of the class $Y(u_\lambda) \in L^0(\Omega)$.

Every $e \in E$ has a unique finite expansion $e = \sum_{\lambda \in F} a_\lambda u_\lambda$. Define

$$X(\omega)(e) = \sum_{\lambda \in F} a_\lambda y_\lambda(\omega). \quad (2)$$

For every ω , the map $e \mapsto X(\omega)(e)$ is linear, so $X(\omega) \in E^*$. For every fixed e , the right side of (2) is a finite sum of measurable functions. Hence $X : \Omega \rightarrow E^*$ is measurable for the cylinder sigma-algebra.

We can summarize the mapping properties as follow.

- (a) As was shown in Sections 3 and 4 the maps κ and κ_0 are always surjective.
- (b) The maps κ , κ_0 , κ_1 , κ_2 and $\bar{\kappa}$ need not be injective (Examples 7.1 7.2 7.3).
- (c) The maps ι and ι_0 need not be surjective (Example 7.4).
- (d) The maps $\bar{\kappa}$, ι_1 , ι_2 and κ_2 are not necessarily surjective (Examples 7.5 and 7.7, Example 7.6, Example 7.8).

It remains open whether κ_1 is surjective, in general. If a locally convex space has property $(\mathcal{R})$, then $\bar{\kappa}$ as well as κ_2 and ι_2 are bijective, and κ_1 is surjective (Remark 5.3).

Figure 1: The open surjectivity question in arXiv:2402.04926, p. 17.

Linearity of Y in $L^0(\Omega)$ gives, for every finite expansion,

$$[X(\cdot)(e)] = \sum_{\lambda \in F} a_\lambda [y_\lambda] = \sum_{\lambda \in F} a_\lambda Y(u_\lambda) = Y(e).$$

Thus $\kappa_0 X = Y$.

It remains only to check that this unrestricted lift lies in the domain of κ_1 . If $e_n \rightarrow e$ in E , sequential continuity of Y gives

$$Y(e_n) \longrightarrow Y(e) \quad \text{in } L^0(\Omega),$$

which is precisely convergence in probability. Since $[X(\cdot)(e_n)] = Y(e_n)$ and $[X(\cdot)(e)] = Y(e)$, we obtain

$$X(\cdot)(e_n) \longrightarrow X(\cdot)(e) \quad \text{in probability.}$$

Therefore $X \in L^0(\Omega : E^*)_{\text{prob}}$, and on this domain $\kappa_1 X = \kappa_0 X = Y$. □

The mechanism can be summarized by the exact identity

$$L^0(\Omega : E^*)_{\text{prob}} = \kappa_0^{-1}(\mathcal{L}(E : L^0(\Omega))_{\text{prob}}). \quad (3)$$

Consequently, surjectivity of κ_0 onto the unrestricted right-hand space implies surjectivity of its restriction onto the probability level.

Lemma 2 (null-set audit). *No common exceptional set is needed in the final continuity step.*

Proof. Convergence in probability depends only on the classes in $L^0(\Omega)$. For any fixed convergent sequence (e_n) , the equalities $[X(\cdot)(e_n)] = Y(e_n)$ involve only countably many coordinates; changing any representative on a null set does not change their convergence in probability. The pathwise common-null-set requirement belongs to the strictly stronger “path” level and is absent from the definition of the domain in (1). □

Corollary 3. *The open question quoted in Figure 1 has an affirmative answer, without assuming property $(\mathcal{R})$.*

Scope. Theorem 1 settles surjectivity of κ_1 . It makes no claim that κ_1 is injective, nor that the maps involving the sequentially continuous dual E' are surjective; the counterexamples to those stronger properties in the source remain untouched.

References

- [1] M. Oberguggenberger, *Generalized stochastic processes revisited*, J. Math. Anal. Appl. 545 (2025), 129172; arXiv:2402.04926.